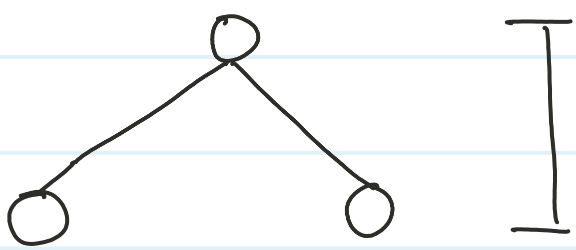
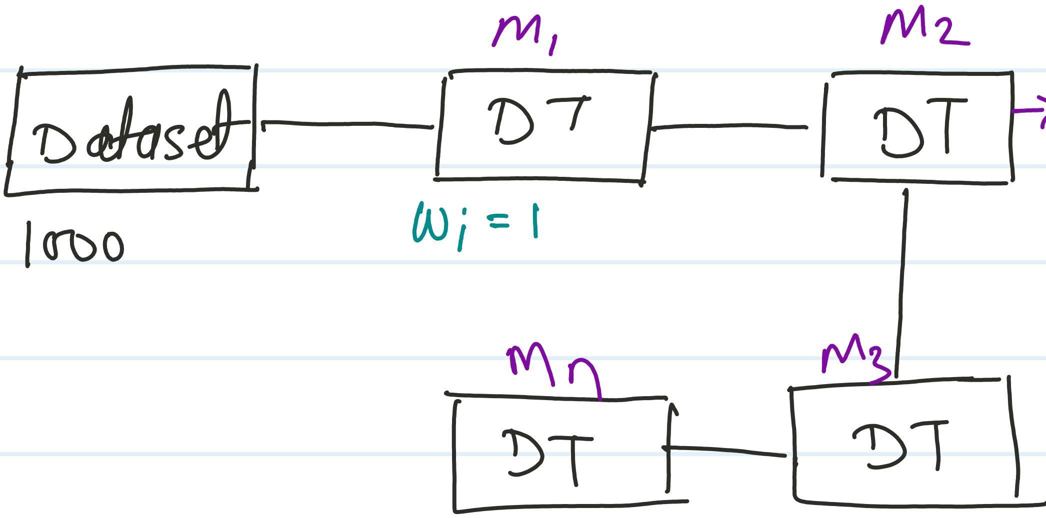
Ada Boost

Same as bagging, Boosting also has homogeneous and heterogeneous model but we use DT only in the ada boosting model.



1 branch of tree called "STUMP"

It is sequancial learning model in this model, every model is known as weaklearner, it pass o/p to the next weaklearner with some weight assigned to it.



Add more weight to misclassified point and reduce for correct class. point

Same process occurs for N numbers of weaklearner based of Combine prediction using a weight majority vote.

AdaBoost commonly used for "classification" problems and very less use for Regression problem.

$$\text{Model} = \alpha_1 (m_1) + \alpha_2 (m_2) + \alpha_3 (m_3) + \cdots + \alpha_n (m_n)$$

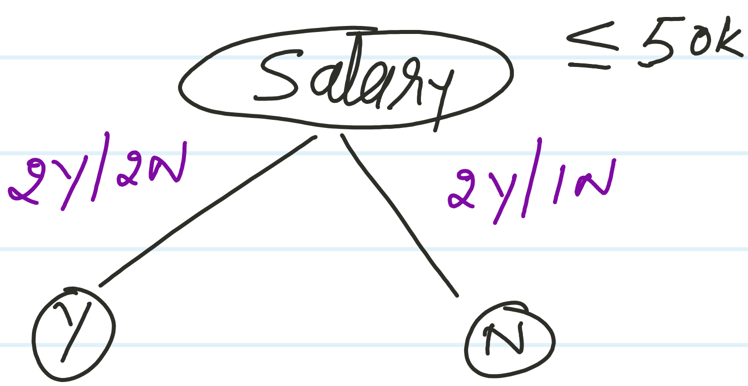
α = weight

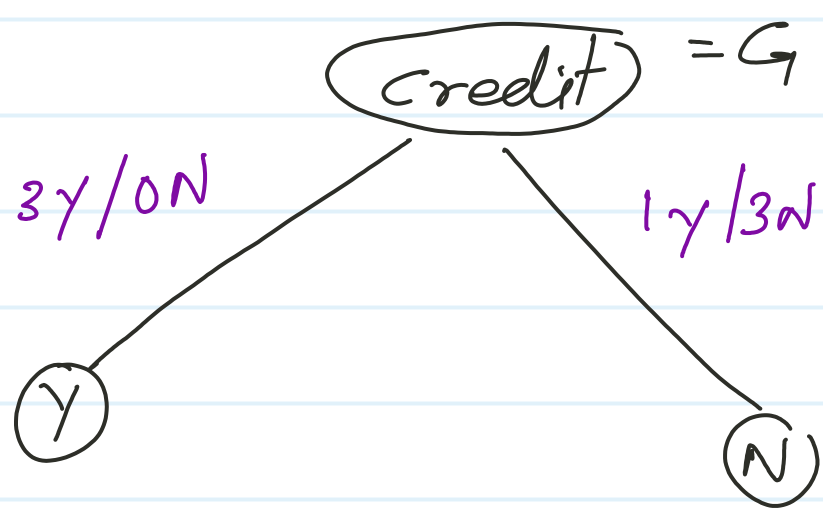
m_1, m_2, m_3, m_n = weaklearner.

Dataset

	Salary	credit	Approach		weight Ass
1	≤ 50K	Bad	No	wi=1	1/7
2	≤ 50K	Good	Yes		1/7
3	≤ 50K	Good	Yes		1/7
4	> 50K	Bad	N		1/7
5	> 50K	Good	Yes		1/7
6	> 50K	Neutral	yes		1/7
7	<= 50K	Neutral	NO		1/7

step-1 we create decision tree stump





We use either Entrop or Gini and Find Information Gain

step - 2 After first weaklearner training, how many wrong predicted point identify.

Add more weight to the wrong predicted point (misclassified point)

Reduce weight to the correct classified point.

step - 3 one wrongly predicted point

$$\text{performance of stump} = \frac{1}{2} \ln \left[\frac{1 - T.E}{T.E} \right]$$

$$\alpha_1 = \frac{1}{2} \ln \left[\frac{1 - 1/7}{1/7} \right]$$

$$\approx 0.896$$

$\alpha_1 = 0.896$

step-4 update weight

For correctly classified point

$= \text{weight} \times e^{-\text{performance of stump}}$

$= \frac{1}{7} \times e^{-(0.896)}$

$= 0.058$

For incorrect classified point

$= \text{weight} \times e^{\text{performance of stump}}$

$= 1/7 \times e^{(0.896)}$

$= 0.349$

For corret data point weight decrease

For incorrect datapoint weight Incrase.

Salary	Credit	Approch	predicted point	updated weight
—	—	Y	Y	0.058
—	—	N	N	0.058
—	—	Y	-N	0.349
—	—	N	N	0.058
—	—	Y	Y	0.058
—	—	Y	-N	0.349
—	—	Y	Y	0.058

Step - 5

Normalization weight computation and assign Bins

Sal	credi.	App.	weight	update weight	Normalization
—	—	—	—	0.058 ÷ 0.988	0.0587
—	—	—	—	0.058 ÷ 0.988	0 0587
—	—	—	—	0.349 ÷ 0 988	0.353
—	—	—	—	0.058 ÷ 0 988	0.0587
—	—	—	—	0.058 ÷ 0 988	0.0587
—	—	—	—	0.349 ÷ 0.988	0.353
				0.058 ÷ 0.988	0.0587
				0.988	1